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Summary

Java Backend Developer (Computer Engineering, 2024) with 3 end-to-end Spring Boot projects — including a **live, AWS-deployed multi-tenant AI SaaS product** with **RAG and agentic MCP tool-calling**. Hands-on with REST API design, JWT security, concurrency-safe transactions, PostgreSQL schema modeling, and Docker-based deployment. Strong in OOP, DSA (200+ problems in Java), debugging, and root-cause analysis.

Technical Skills

Languages: Java (8–21), SQL

Backend & Frameworks: Spring Boot, Spring MVC, Spring Data JPA, Hibernate, REST API Design, JWT Authentication, Maven

Databases: PostgreSQL, Oracle, Relational Schema Design, JPA/Hibernate ORM, Transactions & Concurrency Control

Cloud & DevOps: AWS (Elastic Beanstalk, RDS, ECS), Docker, Docker Compose, Linux, Git, GitHub Actions, Postman

Generative AI (Basics): Spring AI, Model Context Protocol (MCP), LLM Integration

Core CS: Data Structures & Algorithms, OOP, Low-Level Design, Debugging & Root Cause Analysis, Complexity Analysis, DBMS & OS fundamentals

Projects

ComplianceIQ — AI Payroll Compliance SaaS Product for CA Firms | *Java 21, Spring Boot, Spring AI, RAG, PGVector, MCP, PostgreSQL, React, Docker, AWS* [Live Demo](#) | [Source Code](#)

- Architected and shipped a **live multi-tenant SaaS product on AWS EC2** automating Indian payroll compliance (EPF, ESI, TDS, PT) across **30+ REST APIs** — cutting a CA's 2–3 hour Excel workflow to **under 2 minutes**.
- Engineered an **agentic AI assistant** (Spring AI + MCP): the LLM autonomously invokes **6 server-side tools** to run live compliance checks, surface violations with fixes, and generate downloadable audit reports.
- Built a **RAG pipeline** on PGVector (384-dim MiniLM embeddings, HNSW cosine search) serving **citation-backed answers** from law documents, with an instant knowledge-update API — new rules live for all tenants in seconds, **zero redeployment**.
- Secured the platform with **stateless JWT, BCrypt, per-IP rate limiting (Bucket4j)** and tenant-scoped isolation across 7 PostgreSQL entities; automated Form 16/ECR/challans, AI-summarized audit PDFs, bulk payslip ZIPs, and scheduled deadline email alerts.
- Containerized via **multi-stage Docker builds (450 MB → 250 MB)** with an Nginx-proxied React frontend; resolved a production **OOM crash via JVM heap tuning and swap provisioning** for stable 24/7 uptime on AWS EC2.

Global Class Offering Booking System | *Java 21, Spring Boot, PostgreSQL, Docker, REST APIs* [Source Code](#)

- Designed and built a scalable backend managing the full lifecycle of course offerings, multi-timezone scheduling, and booking workflows for a global education platform across 15+ REST endpoints.
- Improved reliability of concurrent bookings by implementing conflict detection and transactional integrity with Spring @Transactional, preventing double-booking under simultaneous requests.
- Engineered a timezone-aware scheduling engine using the Java Time API (Instant, ZoneId), eliminating cross-region time-conversion errors for users in different geographies.
- Modeled a normalized PostgreSQL schema (5+ entities) with JPA/Hibernate and optimized foreign-key relationships; containerized with Docker Compose and validated all endpoints end-to-end via Postman.

AI Ticketing System — Helpdesk Backend | *Java 21, Spring Boot, Spring AI, PostgreSQL* [Source Code](#)

- Engineered a self-service helpdesk backend operating the complete ticket lifecycle — creation, automatic priority detection (Low/Medium/High), team routing, and resolution — modeled on incident-management workflows.
- Designed a rule-based classification engine with a pluggable service layer, enabling drop-in LLM/Generative AI classification without refactoring core services.
- Reduced debugging and root-cause-analysis time by implementing centralized global exception handling and the DTO pattern, producing consistent, observable error responses across all APIs.
- Modeled the PostgreSQL schema with Spring Data JPA across users, tickets, and teams; validated create, assign, prioritize, and resolve flows via automated Postman test runs.

Education

B.E. in Computer Engineering — Mumbai University

June 2020 – May 2024

Certifications & Problem Solving

Certifications: [Java Full Stack Developer](#) | [DSA with Java](#) | [Udemy Java Programming](#)

Problem Solving: 200+ DSA problems solved in Java on [LeetCode](#) (arrays, strings, trees, graphs, dynamic programming)